WORKED SOLUTION

STEP III 2023, Question 5

Pure • Number Theory

STEP III 2023, Question 5

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Each part turns an equation in integers into a *product* equal to something whose factors we know completely: 49, then pq, then $3p^2q$. Once one side is a product of two integers, the size of each factor (which one is bigger, which is smaller than p) cuts the list of possibilities down to one or two cases. In part (iii) the factor we control is $p+q-n$, and the inequalities we are asked to prove are exactly what stop it sharing a factor with p or q.

Solution

Part (i)

Multiplying through by $7xy$ (both non-zero) gives

$$7y+14x=2xy \quad\Longleftrightarrow\quad 2xy-14x-7y=0.$$

Since $(2x-7)(y-7)=2xy-14x-7y+49$, adding 49 to both sides gives $(2x-7)(y-7)=49$, as required.

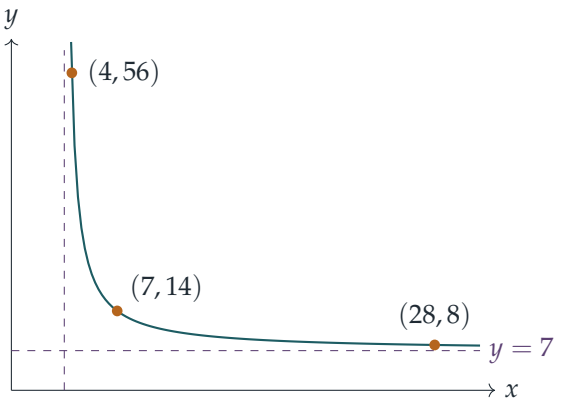
Now let x and y be positive integers. Then $2x-7$ and $y-7$ are integers whose product is 49, so they are a factor pair of 49:

$$(2x-7,\ y-7)\in\{(1,49),\,(7,7),\,(49,1),\,(-1,-49),\,(-7,-7),\,(-49,-1)\}.$$

Positivity rules out the negative pairs: $x\geq 1$ forces $2x-7\geq -5$, and $y\geq 1$ forces $y-7\geq -6$, so neither factor can be -7 or -49, and every negative pair contains one of these. The positive pairs give

$$(x,y)=(4,56),\quad (7,14),\quad (28,8),$$

and each does satisfy the original equation: for example $\frac{1}{4}+\frac{2}{56}=\frac{7}{28}+\frac{1}{28}=\frac{2}{7}$.



The curve is a rectangular hyperbola with asymptotes $x = \frac{7}{2}$ and $y = 7$; the three solutions are the lattice points on its positive branch.

Part (ii)

Expanding $(p+q)^2 = p^2 + 2pq + q^2 = n^2 + pq$, so

$$(p+q+n)(p+q-n) = (p+q)^2 - n^2 = pq.$$

The right-hand side is positive and $p+q+n > 0$, so $p+q-n$ is a *positive* integer. The positive divisors of pq are $1, p, q, pq$ (just $1, p, p^2$ if $p = q$), and $p+q+n$ is one of them. But $n > 0$ means $p+q+n > p$ and $p+q+n > q$, so $p+q+n$ is neither p nor q; the only divisor left is pq. Hence $p+q+n = pq$ and the co-factor is

$$p+q-n = 1, \qquad \text{that is,} \qquad p+q = n+1.$$

Substituting $n = p+q-1$ into $p+q+n = pq$ gives $2p+2q-1 = pq$, which rearranges to

$$pq - 2p - 2q + 4 = 3 \quad \Longleftrightarrow \quad (p-2)(q-2) = 3.$$

Primes satisfy $p-2 \geq 0$, so both factors are non-negative, and the product is 3, so $\{p-2, q-2\} = \{1, 3\}$. Therefore

$$(p,q) = (3,5) \quad \text{or} \quad (5,3),$$

and indeed $9 + 15 + 25 = 49 = 7^2$, with $n = 7 = 3 + 5 - 1$.

Part (iii)

Here n is the positive real cube root of $p^3 + q^3 + 3pq^2$.

The inequalities. Since $p, q > 0$,

$$n^3 = q^3 + (p^3 + 3pq^2) > q^3, \qquad n^3 = p^3 + (q^3 + 3pq^2) > p^3,$$

and cubing is strictly increasing, so $n > q$ and $n > p$. These rearrange to $p + q - n < p$ and $p + q - n < q$ respectively. We shall also need the lower bound:

$$(p+q)^3 = p^3 + 3p^2q + 3pq^2 + q^3 = n^3 + 3p^2q > n^3,$$

so $p + q > n$, i.e. $p + q - n > 0$.

No prime solutions. Suppose p and q are primes and n is an integer. Put $d = p + q - n$, an integer with

$$0 < d < p, \qquad 0 < d < q.$$

From the expansion above, $(p+q)^3 - n^3 = 3p^2q$, and factorising a difference of cubes,

$$d\left((p+q)^2 + (p+q)n + n^2\right) = 3p^2q,$$

so d divides $3p^2q$.

? **Ask yourself** Why can d not share a factor with p or q?

Because p is prime, the only way for d to share a factor with p is for p to divide d, which would need $d \geq p$. But $d < p$. In the same way $d < q$ means q does not divide d. So d is coprime to p^2q, and $d \mid 3p^2q$ forces $d \mid 3$. This is exactly what the inequalities were for. Hence $d = 1$ or $d = 3$. Write $s = p + q$.

Case $d = 1$. Then $n = s - 1$ and

$$3p^2q = s^3 - (s-1)^3 = 3s^2 - 3s + 1,$$

which is impossible: the left-hand side is a multiple of 3 and the right-hand side leaves remainder 1 on division by 3.

Case $d = 3$. Now $3 = d < p$ and $3 < q$, so neither p nor q is 3. Here $n = s - 3$ and

$$3p^2q = s^3 - (s-3)^3 = 9s^2 - 27s + 27, \qquad \text{so} \qquad p^2q = 3(s^2 - 3s + 3).$$

Thus $3 \mid p^2q$, and since 3 is prime it divides p or q, making one of them equal to 3. That contradicts $p, q > 3$.

Both cases are impossible, so there are no primes p and q for which $p^3 + q^3 + 3pq^2$ is the cube of an integer. ■

Checks

A brute-force search over all positive integers $x, y < 2000$ finds exactly the three pairs in (i). Over all primes $p, q < 1500$, $p^2 + pq + q^2$ is a perfect square only for $\{p, q\} = \{3, 5\}$ (giving 49), and $p^3 + q^3 + 3pq^2$ is never a perfect cube. Random positive reals confirm $0 < p + q - n < \min(p, q)$ in (iii). A quick hand check of (ii): $(3 + 5 + 7)(3 + 5 - 7) = 15 \times 1 = 15 = pq$.

Common slips

- Forgetting the negative factor pairs of 49 in (i). They give no solutions here, but they have to be ruled out, not ignored.
- In (ii), assuming without justification that $p+q-n$ is the smaller factor and positive: both follow from $n>0$ and $pq>0$, and they need saying.
- In (iii), stopping at $d \mid 3$. The two cases $d=1$ and $d=3$ still have to be eliminated, and the second needs the fact that $d<p,q$ once more.

Beyond STEP

Every part uses the same trick: rewrite a Diophantine equation as a product of integers equal to a number whose factorisation you know, then use size and coprimality to pin the factors down. Part (i) belongs to the family of Egyptian-fraction equations $\frac{1}{x}+\frac{1}{y}=\frac{1}{n}$, and part (ii) is about integer triangles with a 120° angle, since $p^2+pq+q^2=n^2$ is the cosine rule for that angle. Davenport's *The Higher Arithmetic* is a lovely introduction to this kind of elementary number theory.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.